

CHAPTER C21

SITE-SPECIFIC GROUND MOTION PROCEDURES FOR SEISMIC DESIGN

C21.0 GENERAL

Site-specific procedures for computing earthquake ground motions include dynamic site response analyses and probabilistic and deterministic seismic hazard analyses (PSHA and DSHA), which may include dynamic site response analysis as part of the calculation. Use of site-specific procedures may be required in lieu of the general procedure in Sections 11.4.2 through 11.4.7; Section C11.4.8 in 7-16 explains the conditions under which the use of these procedures is required. Such studies must be comprehensive and must incorporate current scientific interpretations. Because there is typically more than one scientifically credible alternative for models and parameter values used to characterize seismic sources and ground motions, it is important to formally incorporate these uncertainties in a site-specific analysis. For example, uncertainties may exist in seismic source location, extent, and geometry; maximum earthquake magnitude; earthquake recurrence rate; ground motion attenuation; local site conditions, including soil layering and dynamic soil properties; and possible two- or three-dimensional wave-propagation effects. The use of peer review for a site-specific ground motion analysis is encouraged.

Site-specific ground motion analysis can consist of one of the following approaches: (a) PSHA and possibly DSHA if the site is near an active fault, (b) PSHA/DSHA followed by dynamic site response analysis, and (c) dynamic site response analysis only. The first approach is used to compute ground motions for bedrock or stiff soil conditions (not softer than Site Class D). In this approach, if the site consists of stiff soil overlying bedrock, for example, the analyst has the option of either (a) computing the bedrock motion from the PSHA/DSHA and then using the site coefficient (F_a and F_v) tables in Section 11.4.3 to adjust for the stiff soil overburden or (b) computing the response spectrum at the ground surface directly from the PSHA/DSHA. The latter requires the use of attenuation equations for computing stiff soil-site response spectra (instead of bedrock response spectra).

The second approach is used where softer soils overlie the bedrock or stiff soils. The third approach assumes that a site-specific PSHA/DSHA is not necessary but that a dynamic site response analysis should or must be performed. This analysis requires the definition of an outcrop ground motion, which can be based on the 5% damped response spectrum computed from the PSHA/DSHA or obtained from the general procedure in Section 11.4. A representative set of acceleration time histories is selected and scaled to be compatible with this outcrop spectrum. Dynamic site response analyses using these acceleration histories as input are used to compute motions at the ground surface. The response spectra of these surface motions are used to define a maximum considered earthquake (MCE) ground motion response spectrum.

The approaches described in the aforementioned have advantages and disadvantages. In many cases, user preference governs the selection, but geotechnical conditions at the site may dictate the use of one approach over the other. If bedrock is at a depth much greater than the extent of the site geotechnical investigations, the direct approach of computing the ground surface motion in the PSHA/DSHA may be more reasonable. On the other hand, if bedrock is shallow and a large impedance contrast exists between it and the overlying soil (i.e., density times shear wave velocity of bedrock is much greater than that of the soil), the two-step approach might be more appropriate.

Use of peak ground acceleration as the anchor for a generalized site-dependent response spectrum is discouraged because sufficiently robust ground motion attenuation relations are available for computing response spectra in western U.S. and eastern U.S. tectonic environments.

C21.1 SITE RESPONSE ANALYSIS

C21.1.1 Base Ground Motions. Ground motion acceleration histories that are representative of horizontal rock motions at the site are required as input to the soil model. Where a site-specific ground motion hazard analysis is not performed, the MCE response spectrum for Site Class B (rock) is defined using the general procedure described in Section 11.4.1. If the model is terminated in material of Site Class A, C, or D, the input MCE response spectrum is adjusted in accordance with Section 11.4.3. The U.S. Geological Survey (USGS) national seismic hazard mapping project website (earthquake.usgs.gov/hazards/products/conterminous) includes hazard deaggregation options that can be used to evaluate the predominant types of earthquake sources, magnitudes, and distances contributing to the probabilistic ground motion hazard. Sources of recorded acceleration time histories include the databases of the Consortium of Organizations for Strong Motion Observation Systems (COSMOS) Virtual Data Center website (www.cosmos-eq.org), the Pacific Earthquake Engineering Research (PEER) Center Strong Motion Database website (peer.berkeley.edu/products/strong_ground_motion_db.html), and the U.S. National Center for Engineering Strong Motion Data (NCESMD) website (<http://www.strongmotioncenter.org>). Ground motion acceleration histories at these sites generally were recorded at the ground surface and hence apply for an outcropping condition and should be specified as such in the input to the site response analysis code (Kwok et al. 2007 have additional details).

C21.1.2 Site Condition Modeling. Modeling criteria are established by site-specific geotechnical investigations that should include (a) borings with sampling; (b) standard penetration tests (SPTs), cone penetrometer tests (CPTs), and/or other subsurface investigative techniques; and (c) laboratory

testing to establish the soil types, properties, and layering. The depth to rock or stiff soil material should be established from these investigations. Investigation should extend to bedrock or, for very deep soil profiles, to material in which the model is terminated. Although it is preferable to measure shear wave velocities in all soil layers, it is also possible to estimate shear wave velocities based on measurements available for similar soils in the local area or through correlations with soil types and properties. A number of such correlations are summarized by Kramer (1996).

Typically, a one-dimensional soil column extending from the ground surface to bedrock is adequate to capture first-order site response characteristics. For very deep soils, the model of the soil columns may extend to very stiff or very dense soils at depth in the column. Two- or three-dimensional models should be considered for critical projects when two- or three-dimensional wave propagation effects may be significant (for example, sloping ground sites). The soil layers in a one-dimensional model are characterized by their total unit weights and shear wave velocities from which low-strain (maximum) shear moduli may be obtained and by relationships defining the nonlinear shear stress-strain behavior of the soils. The required relationships for analysis are often in the form of curves that describe the variation of soil shear modulus with shear strain (modulus reduction curves) and by curves that describe the variation of soil damping with shear strain (damping curves). In a two- or three-dimensional model, compression wave velocities or moduli or Poisson ratios also are required. In an analysis to estimate the effects of liquefaction on soil site response, the nonlinear soil model also must incorporate the buildup of soil pore water pressures and the consequent reductions of soil stiffness and strength. Typically, modulus reduction curves and damping curves are selected on the basis of published relationships for similar soils (for example, Vucetic and Dobry 1991, Electric Power Research Institute 1993, Darendeli 2001, Menq 2003, and Zhang et al. 2005). Site-specific laboratory dynamic tests on soil samples to establish nonlinear soil characteristics can be considered where published relationships are judged to be inadequate for the types of soils present at the site. Shear and compression wave velocities and associated maximum moduli should be selected based on field tests to determine these parameters or, if such tests are not possible, on published relationships and experience for similar soils in the local area. The uncertainty in the selected maximum shear moduli, modulus reduction and damping curves, and other soil properties should be estimated (Darendeli 2001, Zhang et al. 2008). Consideration of the ranges of stiffness prescribed in Section 12.13.3 (increasing and decreasing by 50%) is recommended.

C21.1.3 Site Response Analysis and Computed Results.

Analytical methods may be equivalently linear or nonlinear. Frequently used computer programs for one-dimensional analysis include the equivalent linear program SHAKE (Schnabel et al. 1972, Idriss and Sun 1992) and the nonlinear programs FLAC (Itasca 2005); DESRA-2 (Lee and Finn 1978); MARDES (Chang et al. 1991); SUMDES (Li et al. 1992); D-MOD_2 (Matasovic 2006); DEEPSOIL (Hashash and Park 2001); TESS (Pyke 2000); and OpenSees (Ragheb 1994, Parra 1996, and Yang 2000). If the soil response induces large strains in the soil (such as for high acceleration levels and soft soils), nonlinear programs may be preferable to equivalent linear programs. For analysis of liquefaction effects on site response, computer programs that incorporate pore water pressure development (effective stress analyses) should be used (for example, FLAC, DESRA-2, SUMDES, D-MOD_2, TESS, DEEPSOIL, and OpenSees). Response spectra of output motions at the ground surface are

calculated as the ratios of response spectra of ground surface motions to input outcropping rock motions. Typically, an average of the response spectral ratio curves is obtained and multiplied by the input MCE response spectrum to obtain the MCE ground surface response spectrum. Alternatively, the results of site response analyses can be used as part of the PSHA using procedures described by Goulet et al. (2007) and programmed for use in OpenSHA (www.opensha.org; Field et al. 2005). Sensitivity analyses to evaluate effects of soil-property uncertainties should be conducted and considered in developing the final MCE response spectrum.

C21.2 RISK-TARGETED MAXIMUM CONSIDERED EARTHQUAKE (MCE_R) GROUND MOTION HAZARD ANALYSIS

Site-specific risk-targeted maximum considered earthquake (MCE_R) ground motions are based on separate calculations of site-specific probabilistic and site-specific deterministic ground motions.

Both the probabilistic and deterministic ground motions are defined in terms of 5% damped spectral response in the maximum direction of horizontal response. The maximum direction in the horizontal plane is considered the appropriate ground motion intensity parameter for seismic design using the equivalent lateral force (ELF) procedure of Section 12.8 with the primary intent of avoiding collapse of the structural system.

Most ground motion relations are defined in terms of average (geometric mean) horizontal response. Maximum response in the horizontal plane is greater than average response by an amount that varies with period. Maximum response may be reasonably estimated by factoring average response by period-dependent factors, such as 1.1 at short periods and 1.3 at a period of 1.0 s (Huang et al. 2008). The maximum direction was adopted as the ground motion intensity parameter for use in seismic design in lieu of explicit consideration of directional effects.

C21.2.1 Probabilistic (MCE_R) Ground Motions.

Probabilistic seismic hazard analysis (PSHA) methods and subsequent computations of risk-targeted probabilistic ground motions based on the output of PSHA are sufficient to define MCE_R ground motion at all locations except those near highly active faults. Descriptions of current PSHA methods can be found in McGuire (2004). The primary output of PSHA methods is a so-called hazard curve, which provides mean annual frequencies of exceeding various user-specified ground motion amplitudes. Risk-targeted probabilistic ground motions are derived from hazard curves using one (or both for comparison purposes) of the methods described in the following two subsections.

C21.2.1.1 Method 1. The simpler but more approximate method of computing a risk-targeted probabilistic ground motion for each spectral period in a response spectrum is to first interpolate from a site-specific hazard curve the ground motion for a mean annual frequency corresponding to 2% probability of exceedance in 50 years (namely 1/2,475 per year). Then this “uniform-hazard” ground motion is factored by a so-called risk coefficient for the site location that is based on those mapped in Figs. 22-18 and 22-19. Via the method explained in the next subsection, the mapped risk coefficients have been computed from the USGS hazard curves for Site Class B and spectral periods of 0.2 and 1.0 s.

C21.2.1.2 Method 2. The direct method of computing risk-targeted probabilistic ground motions uses the entire site-specific hazard curve that results from PSHA. The computation is

detailed in Luco et al. (2007). Summarizing, the hazard curve is combined with a collapse fragility (or probability distribution of the ground motion amplitude that causes collapse) that depends on the risk-targeted probabilistic ground motion itself. The combination quantifies the risk of collapse. Iteratively, the risk-targeted probabilistic ground motion is modified until combination of the corresponding collapse fragility with the hazard curve results in a risk of collapse of 1% in 50 years. This target is based on the average collapse risk across the western United States that is expected to result from design for the probabilistic MCE ground motions in ASCE 7.

C21.2.2 Deterministic (MCE_R) Ground Motions.

Deterministic ground motions are to be based on characteristic earthquakes on all known active faults in a region. The magnitude of a characteristic earthquake on a given fault should be a best estimate of the maximum magnitude capable for that fault but not less than the largest magnitude that has occurred historically on the fault. The maximum magnitude should be estimated considering all seismic-geologic evidence for the fault, including fault length and paleoseismic observations. For faults characterized as having more than a single segment, the potential for rupture of multiple segments in a single earthquake should be considered in assessing the characteristic maximum magnitude for the fault.

For consistency, the same attenuation equations and ground motion variability used in the PSHA should be used in the deterministic seismic hazard analysis (DSHA). Adjustments for directivity and/or directional effects should also be made, when appropriate. In some cases, ground motion simulation methods may be appropriate for the estimation of long-period motions at sites in deep sedimentary basins or from great ($M \geq 8$) or giant ($M \geq 9$) earthquakes, for which recorded ground motion data are lacking.

Values of the site coefficients (F_a and F_v) for setting the deterministic (MCE_R) ground motion floor are introduced to incorporate both site amplification and spectrum shape adjustment as described in the research study “Investigation of an Identified Short-Coming in the Seismic Design Procedures of ASCE 7-16 and Development of Recommended Improvements for ASCE 7-16” (Kircher 2015). This study found that the shapes of the response spectra of ground motions were not accurately represented by the shape of the design response spectrum of Figure 11.4-1 for the following site conditions and ground motion intensities: (1) Site Class D where values of $S_1 \geq 0.2$; and (2) Site Class E where values of $S_S \geq 1.0$ and/or $S_1 \geq 0.2$. An adjustment of the corresponding values of F_a and F_v was required to account for this difference in spectrum shape, which was causing the design response spectrum to underestimate long-period motions. Two options were considered to address this shortcoming. For the first option, the subject study developed values of new “spectrum shape adjustment” factors (C_a and C_v) that could be used with site factors (F_a and F_v) to develop appropriate values of design ground motions (S_{DS} and S_{D1}). The second option, ultimately adopted by ASCE 7-16, circumvents the need for these new factors by requiring site-specific analysis for Site Class D site conditions where values of $S_1 \geq 0.2$, and for Site Class E site conditions where values of $S_S \geq 1.0$ and/or $S_1 \geq 0.2$ (i.e., new requirements of Section 11.4.8 of ASCE 7-16). The spectrum shape adjustment factors developed by the subject study for Option 1 provide the basis for the values of site coefficients (F_a and F_v) proposed for Section 21.2.2 and Section 21.3 that incorporate both site amplification and adjustment for spectrum shape. Specifically, the proposed value of $F_v = 2.5$ for Site Class D is based on the product of 1.7 (Site Class D

amplification at $S_1 = 0.6$, without spectrum shape adjustment) and 1.5 (spectrum shape adjustment factor); the proposed value of $F_v = 4.0$ is based on the product of 2.0 (Site Class E amplification at $S_1 = 0.6$ without spectrum shape adjustment) and 2.0 (spectrum shape adjustment factor), where values of spectrum shape adjustment are taken from Section 6.2.2 (Table 11.4-4) of the subject study. The proposed value of $F_a = 1.0$ is based on the product of 0.8 (Site Class E amplification at $S_S = 1.5$ without spectrum shape adjustment) and 1.25 (spectrum shape adjustment factor), where the value of the spectrum shape adjustment is taken from Section 6.2.2 (Table 11.4-3) of the subject study. Site amplification adjusted for spectrum shape effects is approximately independent of ground motion intensity and, for simplicity, the proposed values of site factors adjusted for spectrum shape are assumed to be valid for all ground motion intensities.

C21.2.3 Site-Specific MCE_R . Because of the deterministic lower limit on the MCE_R spectrum (Fig. 21.2-1), the site-specific MCE_R ground motion is equal to the corresponding risk-targeted probabilistic ground motion wherever it is less than the deterministic limit (e.g., 1.5g and 0.6g for 0.2 and 1.0 s, respectively, and Site Class B). Where the probabilistic ground motions are greater than the lower limits, the deterministic ground motions sometimes govern, but only if they are less than their probabilistic counterparts. On the MCE_R ground motion maps in ASCE/SEI 7-10, the deterministic ground motions govern mainly near major faults in California (like the San Andreas) and Nevada. The deterministic ground motions that govern are as small as 40% of their probabilistic counterparts.

C21.3 DESIGN RESPONSE SPECTRUM

Eighty percent of the design response spectrum determined in accordance with Section 11.4.6 was established as the lower limit to prevent the possibility of site-specific studies generating unreasonably low ground motions from potential misapplication of site-specific procedures or misinterpretation or mistakes in the quantification of the basic inputs to these procedures. Even if site-specific studies were correctly performed and resulted in ground motion response spectra less than the 80% lower limit, the uncertainty in the seismic potential and ground motion attenuation across the United States was recognized in setting this limit. Under these circumstances, the allowance of up to a 20% reduction in the design response spectrum based on site-specific studies was considered reasonable.

As described in Section 21.2.2, values of the site coefficients (F_a and F_v) for setting the deterministic (MCE_R) ground motion floor are introduced to incorporate both site amplification and spectrum shape adjustment.

Although the 80% lower limit is reasonable for sites not classified as Site Class F, an exception has been introduced at the end of this section to permit a site class other than E to be used in establishing this limit when a site is classified as F. This revision eliminates the possibility of an overly conservative design spectrum on sites that would normally be classified as Site Class C or D.

C21.4 DESIGN ACCELERATION PARAMETERS

The S_{DS} criteria of Section 21.4 are based on the premise that the value of the parameter S_{DS} should be taken as 90% of peak value of site-specific response spectral acceleration regardless of the period (greater than or equal to 0.2 s) at which the peak value of response spectral acceleration occurs. Consideration of periods

beyond 0.2 s recognizes that site-specific studies (e.g., softer site conditions) can produce response spectra with ordinates at periods greater than 0.2 s that are significantly greater than those at 0.2 s. Periods less than 0.2 s are excluded for consistency with the 0.2-s period definition of the short-period ground motion parameter, S_s , and recognizing that certain sites, such as Central and Eastern United States (CEUS) sites, could have peak response at very short periods that would be inappropriate for defining the value of the parameter S_{DS} . The upper bound limit of 5 s precludes unnecessary checking of response at periods that cannot govern the peak value of site-specific response spectral acceleration. Ninety percent (rather than 100%) of the peak value of site-specific response spectral acceleration is considered appropriate for defining the parameter S_{DS} (and the domain of constant acceleration) since most short-period structures have a design period that is not at or near the period of peak response spectral acceleration. Away from the period of peak response, response spectral accelerations are less, and the domain of constant acceleration is adequately described by 90% of the peak value. For those short-period structures with a design period at or near the period of peak response spectral acceleration, anticipated yielding of the structure during MCE_R ground motions effectively lengthens the period and shifts dynamic response to longer periods at which spectral demand is always less than that at the peak of the spectrum.

The S_{D1} criteria of Section 21.4 are based on the premise that the value of the parameter S_{D1} should be taken as 100% of the peak value of site-specific response spectral acceleration for a period range, $1 \text{ s} \leq T \leq 2 \text{ s}$, for stiffer sites $v_{s,30} \text{ ft/s} > 1,200 \text{ ft/s}$ ($v_{s,30} \text{ m/s} > 366 \text{ m/s}$) similar to the previous requirements of Section 21.4 of ASCE 7-10 and for a period range, $1 \text{ s} \leq T \leq 5 \text{ s}$, for softer sites $v_{s,30} \text{ ft/s} \leq 1,200 \text{ ft/s}$ ($v_{s,30} \text{ m/s} > 366 \text{ m/s}$), which are expected to have peak values of response spectral velocity at periods greater than 2 s. The criteria use the maximum value of the product, TS_a , over the period range of interest to effectively identify the period at which the peak value of response spectral velocity occurs. Consideration of periods beyond 1 s accounts for the possibility that the assumed $1/T$ proportionality for the constant velocity portion of the design response spectrum begins at periods greater than 1 s or is actually $1/T^n$ (where $n < 1$). Periods less than 1 s are excluded for consistency with the definition of the 1-s ground motion parameter, S_1 . Peak velocity response is expected to occur at periods less than or equal to 5 s, and periods beyond 5 s are excluded by the criteria to avoid potential misuse of very long period ground motions that may not be reliable. One hundred percent (rather than a reduced percentage) of the peak value of site-specific response spectral acceleration at the period of peak velocity response is considered appropriate for defining the value of the parameter S_{D1} since response spectral accelerations can be approximately proportional to the assumed $1/T$ shape of the domain of constant velocity for design periods of interest.

C21.5 MAXIMUM CONSIDERED EARTHQUAKE GEOMETRIC MEAN (MCE_g) PEAK GROUND ACCELERATION

Site-specific requirements for determination of peak ground acceleration (PGA) are provided in a new Section 21.5 that is parallel to the procedures for developing site-specific response spectra in Section 21.2. The site-specific MCE peak ground acceleration, PGA_M , is taken as the lesser of the probabilistic geometric mean peak ground acceleration of Section 21.5.1 and the deterministic geometric mean peak ground acceleration of Section 21.5.2. Similar to the provisions for site-specific spectra, a deterministic lower limit

is prescribed for PGA_M with the intent to limit application of deterministic ground motions to the site regions containing active faults where probabilistic ground motions are unreasonably high. However, the deterministic lower limit for PGA_M (in g) is set at a lower value, $0.5 F_{PGA}$, than the value set for the zero-period response spectral acceleration, $0.6 F_a$. The rationale for the value of the lower deterministic limit for spectra is based on the desire to limit minimum spectral values, for structural design purposes, to the values given by the 1997 Uniform Building Code (UBC) for Zone 4 (multiplied by a factor of 1.5 to adjust to the MCE level). This rationale is not applicable to PGA_M for geotechnical applications, and therefore a lower value of $0.5 F_{PGA}$ was selected. Section 21.5.3 of ASCE 7-10 states that the site-specific MCE peak ground acceleration cannot be less than 80% of PGA_M derived from the PGA maps. The 80% limit is a long-standing base for site-specific analyses in recognition of the uncertainties and limitations associated with the various components of a site-specific evaluation.

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